Breast Lesion Detection in Mammography using Deep Learning

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Abstract

Breast cancer detection in mammography is challenging due to the small size and low contrast of lesions such as masses and calcifications. In this work, we evaluate object detection models from different paradigms: one-stage (YOLOv8, RetinaNet), two-stage (Faster R-CNN), and transformer-based (RT-DETR).

The dataset is preprocessed and adapted to standard formats, and all models are fine-tuned on annotated mammographic images. Results show that all approaches struggle with small lesions, especially calcifications, although each paradigm presents different strengths. This study highlights the limitations of current detectors and the need for improved methods in medical imaging.

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1 Introduction

Breast cancer is a major global health concern, accounting for a significant proportion of cancer-related deaths among women. Early detection is crucial for improving patient outcomes, and mammography is the standard imaging technique used in screening programs. Despite its effectiveness, interpreting mammograms remains a complex and time-consuming task, even for experienced radiologists.

One of the main challenges in mammography analysis is the detection of subtle abnormalities such as masses and calcifications. These lesions are often small, exhibit low contrast with surrounding tissue, and may appear differently depending on breast density. Additionally, dense breast tissue can obscure lesions, making detection even more difficult.

In recent years, deep learning methods—particularly convolutional neural networks—have shown remarkable performance in computer vision tasks, including object detection. These methods have the potential to assist radiologists by providing automated and consistent detection of suspicious regions.

More recently, transformer-based architectures have been introduced in object detection, enabling global reasoning over the image and reducing the need for handcrafted components such as anchor boxes. These approaches have shown promising results in natural images and are explored in this work through the RT-DETR model.

In this project, we explore the application of state-of-the-art object detection models to the problem of breast lesion detection in mammography. We focus on comparing different detection paradigms, including one-stage and two-stage approaches, and adapting them to the specific constraints of medical imaging data.

2 Related work

Computer-aided detection (CAD) systems have long been used to assist radiologists in identifying suspicious regions in mammograms. Traditional approaches relied on handcrafted features and classical machine learning algorithms, which often struggled to generalize across datasets.

With the advent of deep learning, convolutional neural networks (CNNs) have significantly improved performance in medical image analysis. In particular, object detection frameworks have been increasingly applied to mammography for identifying lesions.

Object detection models can be broadly categorized into:

- Two-stage detectors, such as Faster R-CNN, which first generate region proposals and then classify them. These models tend to achieve high accuracy but at the cost of computational complexity.
- One-stage detectors, such as YOLO and RetinaNet, which perform detection in a single pass. These models are typically faster and more suitable for real-time applications.
- Transformer-based detectors, such as DETR and its variants (e.g., RT-DETR), which reformulate object detection as a direct set prediction problem without relying on anchor boxes. These models enable global context modeling but often require careful training strategies and sufficient data.

In the context of mammography, challenges such as small object size and class imbalance make model selection and adaptation particularly important.

3 Dataset Description

In this study, we use the Digital Mammography Dataset for Breast Cancer Diagnosis (DMID), which contains mammographic images along with lesion annotations.

The dataset includes images in multiple formats, primarily:

- DICOM (medical imaging standard)
- TIFF (converted images)

Annotations are provided in the form of lesion coordinates and radii, which represent suspicious regions in the breast. These annotations are converted into bounding boxes by defining a rectangular region around each lesion to make them compatible with object detection models.

Two types of lesions are considered:

- Masses
- Calcifications

Other labels present in the dataset, such as normal tissue were excluded, as they do not provide bounding box annotations.

The dataset presents several challenges:

- High resolution images, requiring resizing or tiling strategies
- Small lesion size, especially for calcifications
- Class imbalance, with fewer positive samples compared to background
- Images without annotations, which must be handled carefully during training

These characteristics make the dataset representative of real-world clinical scenarios.

4 Data Preprocessing

A comprehensive preprocessing pipeline was implemented to prepare the dataset for training deep learning models.

4.1 Data extraction

The dataset was initially provided as compressed files, including nested ZIP archives. A custom extraction script was developed to recursively extract all files and organize them into a structured directory.

4.2 Image Processing

Given the presence of multiple formats, a unified image loading strategy was implemented:

- DICOM images were read using specialized libraries and converted to standard image arrays
- All images were normalized and converted to 3-channel RGB format to ensure compatibility with deep learning models
- Images were resized to a fixed resolution (512×512) to balance computational efficiency and detail preservation

4.3 Annotation Conversion (COCO Format)

4.4 Data cleaning

To ensure training stability, several filtering steps were applied:

- Removal of invalid bounding boxes (zero or negative area)
- Filtering of extremely small boxes
- Consistency checks between labels and bounding boxes

These steps help avoid training issues such as NaN losses or incorrect detections.

4.5 Data Splitting

A K-Fold cross-validation strategy (K=10) was implemented to improve robustness and allow for fair model comparison. Each fold contains:

- Training set
- Validation set

This approach ensures that all samples are used for both training and validation across different iterations.

4.6 Conversion to YOLO Format

For training YOLO-based models, the dataset was additionally converted into YOLO format:

- Images organized into train/val folders
- Labels stored as text files with normalized bounding boxes

All images were explicitly converted to RGB to avoid channel mismatches during training.

5 Methodology

5.1 Models Overview

Three object detection models from different paradigms were selected:

5.1.1 YOLOv8 (One-stage detector)

YOLOv8 performs detection in a single forward pass, making it highly efficient. It is well-suited for real-time applications and provides a good balance between speed and accuracy.

5.1.2 Faster R-CNN (Two-stage detector)

Faster R-CNN first generates region proposals using a Region Proposal Network (RPN) and then classifies them. It is generally more accurate but computationally slower.

5.1.3 RetinaNet (One-stage detector with Focal Loss)

RetinaNet addresses class imbalance using Focal Loss, which reduces the impact of easy negatives during training. This makes it particularly suitable for datasets with many background regions, such as mammography.

5.1.4 RT-DETR (Transformer-based detector)

RT-DETR is a real-time object detection model based on transformer architectures. Unlike traditional detectors, it does not rely on anchor boxes or region proposals. Instead, it directly predicts a set of bounding boxes and class labels using attention mechanisms.

This approach enables global reasoning across the entire image, which can be beneficial for complex medical images. However, transformer-based models are typically more data-hungry and may struggle when trained on smaller datasets, such as mammography collections.

5.2 Training Setup

All models were trained using the following configuration:

- Image resolution: 512×512
- Batch size: limited due to hardware constraints
- Optimizer: Adam / SGD (depending on model)
- Pretrained weights: used for transfer learning

- Training epochs: limited due to computational constraints

The experiments were conducted on a system equipped with GPU: NVIDIA GeForce MX330

Due to hardware limitations, training time and model complexity were constrained, influencing the experimental design.

5.3 Data Handling Strategy

Special care was taken to handle dataset-specific challenges:

- Images without annotations were included with empty targets
- Bounding boxes were filtered to remove invalid entries
- Class labels were mapped consistently across all models
- All data was converted to tensor format compatible with PyTorch detection models

The task is formulated as an object detection problem, where the model must both:

- Locate lesions using bounding boxes
- Classify each detected lesion as mass or calcification

6 Experiments and Evaluation

6.1 Evaluation Metrics

Different evaluation metrics are used to assess the performance of the object detection models. The evaluation is divided into two parts, detection metrics and classification metrics. The YOLO model is evaluated using standard metrics provided by the Ultralytics framework.

6.1.1 Detection Metrics

We evaluate the ability of the model to correctly localize lesions in the images, regardless of their class. A predicted bounding box is considered correct if it sufficiently overlaps with a ground truth box. This overlap is measured using the Intersection over Union (IoU).

If the IoU between a predicted box and a ground truth box is greater than a predefined threshold (0.5), the detection is considered a True Positive (TP). Otherwise, it is counted as a False Positive (FP). Ground truth boxes that are not matched with any prediction are considered False Negatives (FN).

From these values, the following metrics are computed:

- Precision: the proportion of predicted boxes that are correct.
- Recall: the proportion of real lesions that are successfully detected.

These metrics focus only on the spatial accuracy of the model (whether the lesion is found), without considering whether the predicted class is correct.

6.1.2 Classification Metrics

Classification metrics evaluate whether the model correctly identifies the type of lesion (mass or calcification). Classification is evaluated for detections that are spatially correct,so, a prediction is only considered for classification if its IoU with a ground truth box is above the threshold.

For each valid detection:

- If the predicted class matches the ground truth class, it is counted as a True Positive (TP).
- If the predicted class is incorrect, it is counted as a False Positive (FP).

Additionally, ground truth lesions that are not detected are counted as False Negatives (FN), since they cannot be classified.

From these values, precision and recall are computed again, but now reflecting both correct detection and correct classification. These metrics measure the overall performance of the model in solving the complete task.

6.1.3 Confusion Matrix

To better understand classification performance, a confusion matrix is also computed. This matrix only includes predictions that have a valid IoU match with ground truth boxes.

The confusion matrix does not include missed detections or false positives without a valid match. Therefore, it provides a detailed view of classification errors, but does not reflect the full detection performance.

6.1.4 Ultralytics Evaluation

For YOLO and RT-DETR evaluation is performed using the validation function provided by the Ultralytics library. This framework computes standard object detection metrics:

- mAP@0.5: mean Average Precision at an IoU threshold of 0.5.
- mAP@0.5:0.95: mean Average Precision averaged over multiple IoU thresholds (from 0.5 to 0.95).

These metrics combine both detection and classification performance into a single measure, taking into account precision and recall across different confidence thresholds.

6.2 Experimental Setup

All experiments were conducted using the same dataset splits and preprocessing pipeline to ensure a fair comparison between models.

The dataset was divided using a K-Fold cross-validation strategy (K=10), although due to computational constraints, experiments were performed on a single fold. Each split consisted of a training set and a validation set.

Images were resized to a fixed resolution of 512×512 pixels. All images were converted to RGB format and normalized before being fed into the models.

For Faster R-CNN and RetinaNet, models were implemented using PyTorch and trained using pretrained weights. Training was performed for a limited number of epochs due to hardware constraints, with a small batch size. The AdamW optimizer was used, with a learning rate of 1×10^{-4}.

YOLO and RT-DETR models were trained and evaluated using the Ultralytics framework, which provides built-in training and validation pipelines. Default training configurations were used, with minor adjustments to adapt to the dataset format.

During evaluation, a confidence threshold was applied to filter predictions, and an IoU threshold of 0.5 was used to determine correct detections. For Faster R-CNN and RetinaNet, a custom evaluation pipeline was implemented to compute detection and classification metrics. For YOLO and RT-DETR, standard metrics such as mAP@0.5 and mAP@0.5:0.95 were obtained directly from the framework.

All experiments were conducted on a system equipped with an NVIDIA GeForce MX330 GPU, which imposed limitations on model size, batch size, and training time.

7 Results

This section presents the experimental results obtained with the Faster R-CNN and YOLO models on the mammography lesion detection task. The evaluation focuses on lesion localization (detection of bounding boxes) and lesion type recognition (mass or calcification).

7.1 Faster R-CNN Results

The Faster R-CNN model was evaluated using the custom evaluation pipeline described in the previous section. We considered detection metrics and classification metrics.

For lesion detection, the model obtained a precision of 0.3846 and a recall of 0.4167. These values indicate that the model was able to detect some lesions correctly, but it also produced false positives and missed several annotated lesions.

For lesion classification, the model obtained a precision of 0.3567 and a recall of 0.3923. This happened because the classification evaluation only considered predictions whose bounding boxes were already correct according to the IoU threshold. Therefore, classification performance was directly limited by the detection stage.

Table 1 summarizes the results.

Table 1: Faster R-CNN evaluation results

Metric	Precision	Recall
Detection	0.3846	0.4167
Classification	0.3567	0.3923

The confusion matrix for classification is shown in Figure 1. Only predictions with valid localization were included.

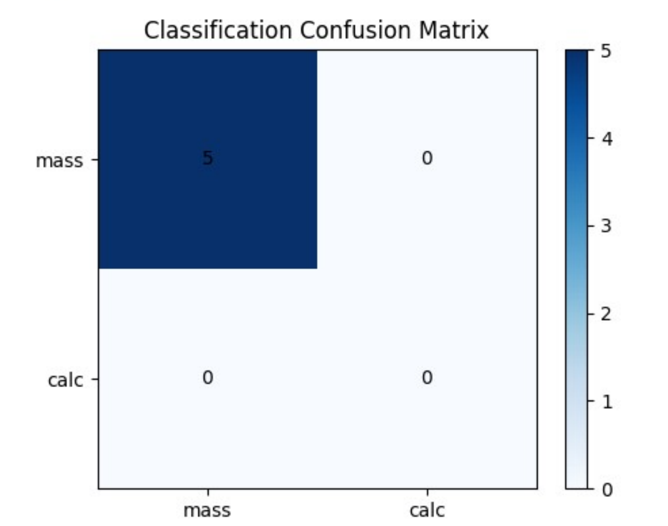


Figure 1: Confusion matrix for Faster R-CNN classification.

This means that five masses were correctly classified, while no calcifications were correctly recognized. In practice, the model showed a clear preference for the mass class and had difficulties learning the calcification category.

The examples also confirmed this behavior. Figure 2 includes three representative predictions:

- One image where a mass was correctly detected with high confidence.
- One image containing a mass and a calcification where no lesions were detected.
- One image containing a mass and a calcification where the model detected the mass but didn't predict the calcification.

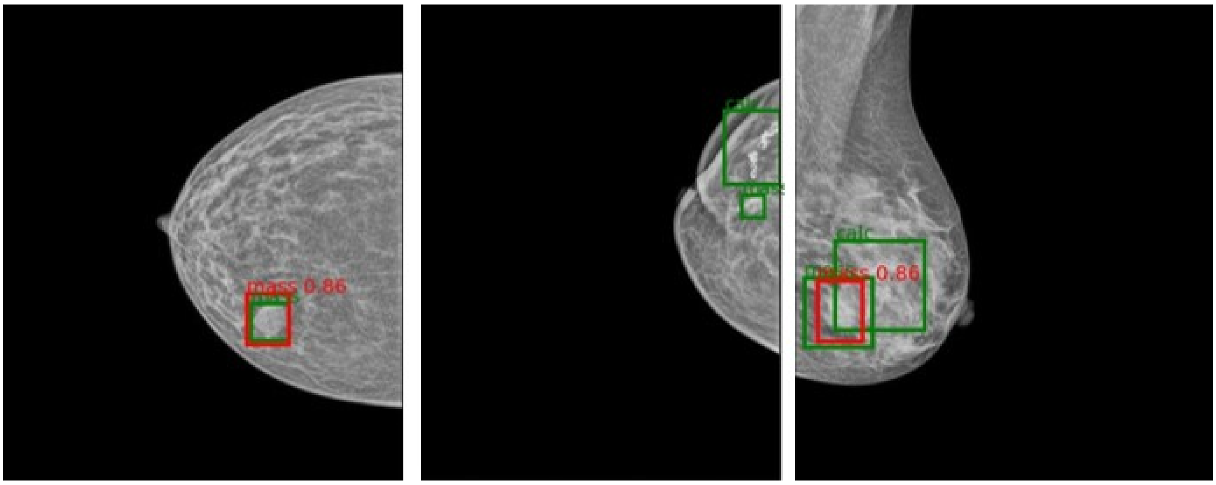


Figure 2: Examples of Faster R-CNN predictions. Green boxes represent ground truth and red boxes represent model predictions.

These results suggest that Faster R-CNN was able to learn the visual appearance of masses better than calcifications. This may be related to the small size and lower contrast of calcifications, which make detection more difficult.

7.2 YOLO Results

The YOLO model was evaluated using the Ultralytics validation framework, which reports mean Average Precision (mAP) and Precision-Recall curves.

Table 2 summarizes these results.

Table 2: YOLO evaluation results

Metric	Value
Mass AP@0.5	0.429
Calcification AP@0.5	0.281
Overall mAP@0.5	0.355

The Precision-Recall curve is included in Figure 3.

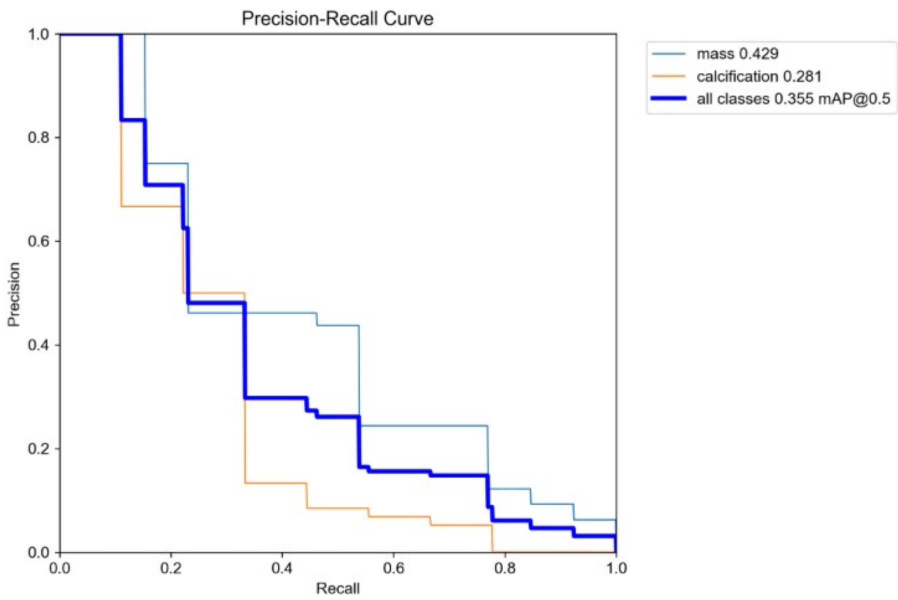
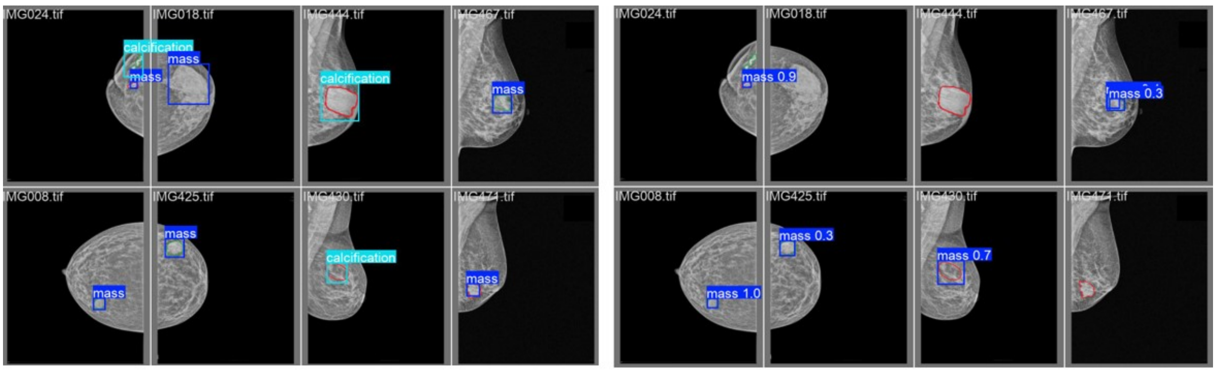


Figure 3: Precision-Recall curve for YOLO.

Visual inspection of prediction examples showed that the model detected some masses correctly, but it also missed several lesions. Similar to Faster R-CNN, calcifications were the most difficult category. In many examples, calcifications were not detected at all.

Figure 4 includes several images with labels and the predictions generated by the model.



Labels Predictions

Figure 4: Examples of YOLO predictions on validation images.

7.3 RetinaNet Results

The RetinaNet model obtained very poor results after training for only three epochs. Due to the limited training time and hardware constraints, the model did not converge properly and was not able to learn meaningful lesion patterns.

Table 3: RetinaNet evaluation results.

Metric	Value
Detection Precision	0.0336
Detection Recall	0.0833
Classification Precision	0.0000
Classification Recall	0.0000

The detection precision was extremely low, which means that most predicted boxes were incorrect. Detection recall was also very low, indicating that only a small number of real lesions were detected.Regarding classification, both precision and recall were zero. This happened because the model did not correctly classify any lesion in the validation set. The confusion matrix contained only zeros, which suggests that no valid predictions were matched with the ground truth boxes under the IoU threshold.

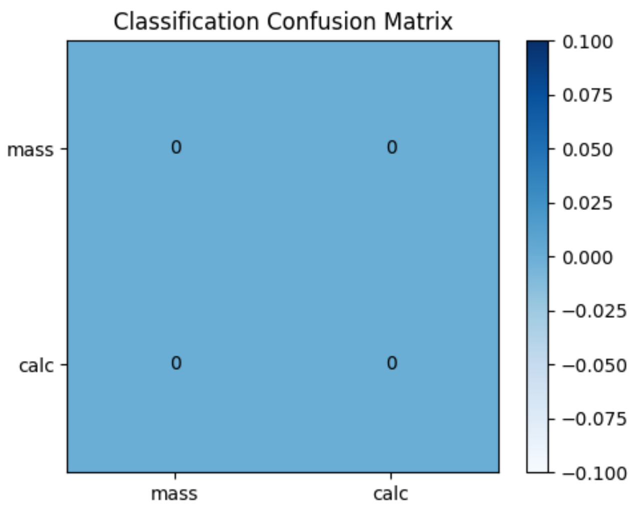


Figure 5: Confusion matrix for RetinaNet classification.

Visual inspection of the predicted images confirms these results.

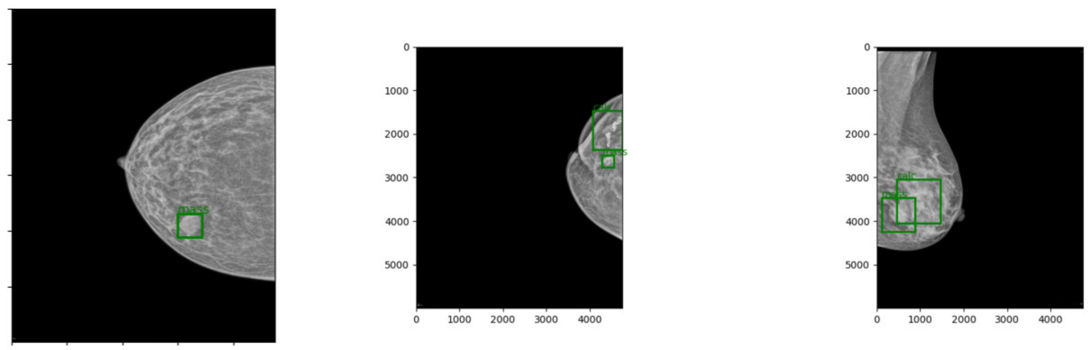


Figure 6: Examples of RetinaNet predictions on validation images. Green boxes represent ground truth and red boxes represent model predictions.

In the examples generated with the visualization function, no red predicted bounding boxes appeared. This indicates that the model was not able to produce confident detections after such a short training process.These results are expected because RetinaNet usually requires more training time to optimize both localization and classification. Training for only three epochs was not enough for the model to learn the characteristics of masses and calcifications. For a fairer comparison, RetinaNet should be trained for more epochs.

7.4 RT-DETR Results

RT-DETR was evaluated on the validation set using the metrics provided by the Ultralytics framework. The model obtained a global mAP@0.5 of 0.184, which indicates limited detection performance on this dataset. Analysing each class separately, the model achieved an AP of 0.367 for *mass*, while the AP for *calcification* was 0.000.

These results suggest that the model was able to detect some masses, but it was not able to learn reliable patterns for calcifications.

Table 4: RT-DETR validation results.

Metric	Value
AP@0.5 (Mass)	0.367
AP@0.5 (Calcification)	0.000
mAP@0.5 (All classes)	0.184

Figure 7 shows the Precision–Recall curve. The curve confirms that the model performed better for the mass class, while the calcification class remained zero.

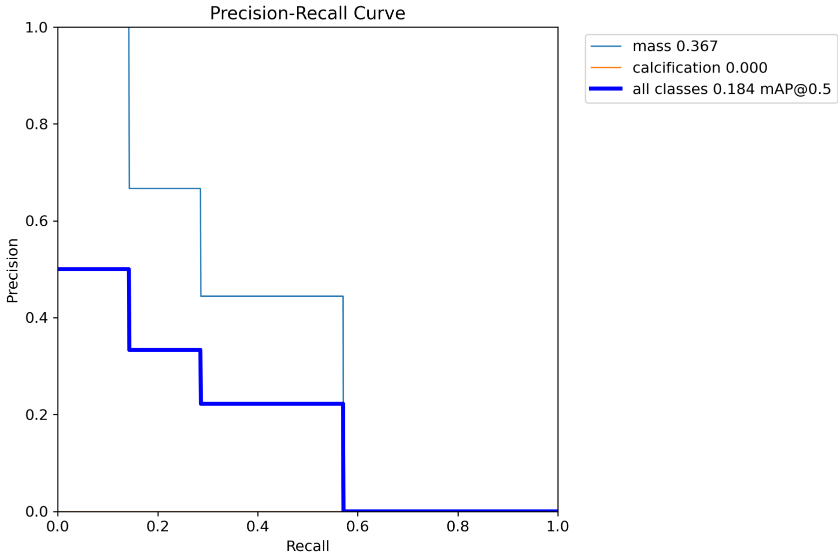


Figure 7: Precision–Recall curve obtained by RT-DETR on the validation set.

Visual inspection of the predictions also revealed several limitations. In some mammograms, the model produced multiple bounding boxes around the same region, generating false positives. In other cases, calcifications were incorrectly classified as masses. Some validation images did not contain any predicted boxes, meaning that the lesion was completely missed.

Figure 8 compares the ground-truth annotations with the predictions generated by RT-DETR. These examples illustrate that, although the model can identify certain suspicious areas, localization and classification remain unstable.

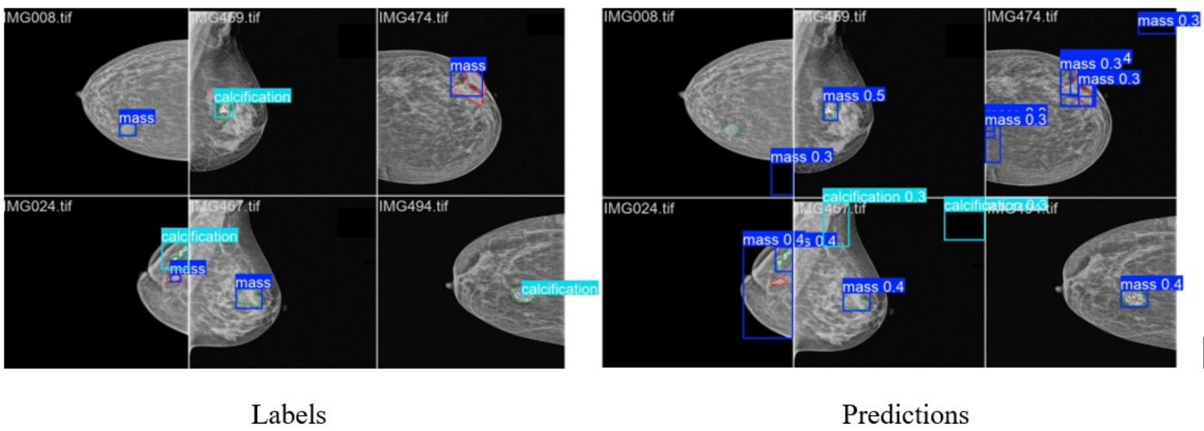


Figure 8: Examples of RT-DETR predictions.

RT-DETR achieved moderate performance for masses but failed to detect calcifications. The model would require more training time and additional data to become competitive on this task.

8 Discussion

The experimental results show that mammographic lesion detection remains a challenging task for all evaluated models. Although some lesions were correctly detected, none of the detectors achieved a consistent performance across both classes. The main difficulty was the detection of calcifications, masses were generally easier to identify. This difference expected because masses are usually larger and have clearer shapes, so models can learn their visual patterns more easily, while calcifications are very small and may appear with low contrast.

YOLO obtained the best overall balance of results. It achieved the highest mAP@0.5 and better class performance than the other approaches. This suggests that one-stage detectors can be a strong option for this task, especially when computational resources are limited. However, even this model still showed weak performance for calcifications, which means that the best model in this study is still far from ideal for clinical use.

Faster R-CNN produced moderate results and showed acceptable recall. It was able to detect several lesions, especially masses, but it also generated false positives and missed many calcifications. As a two-stage detector, Faster R-CNN is usually strong in localisation quality, but in this case the limited dataset size and hardware constraints reduced its potential. With more training time, Faster R-CNN could likely improve significantly.

RetinaNet obtained the weakest results among the CNN-based detectors. In this project, the model was trained for only three epochs due to hardware limitations, and this was not enough for convergence. The absence of correct classifications and the very low detection scores indicate undertraining rather than an intrinsic weakness of the architecture. Therefore, the results obtained here should be interpreted carefully.

RT-DETR achieved intermediate performance. It was able to detect some masses, but it completely failed on calcifications. The model also generated several false positives and unstable predictions. Transformer-based detectors usually require more data and longer training to reach their full potential. The performance of the model may have been limited due to the small dataset and training time. In this project, RT-DETR likely suffered from both factors.

One-stage YOLO model obtained the best global score. The two-stage Faster R-CNN model provided more conservative predictions, although with lower overall performance. RetinaNet was strongly affected by short training. The transformer-based RT-DETR showed potential, but it likely needs more data and computation than available in this work.

Training conditions had a strong influence on the final results. Models were trained for only a small number of epochs because of hardware limitations. Therefore, the comparison reflects not only architectural differences, but also practical constraints. Under a larger computational budget, the ranking between models could change. Faster R-CNN and RT-DETR, in particular, may improve notably with longer training.

Another important limitation is the dataset itself. Medical datasets are often small, imbalanced, and difficult to annotate. This affects generalisation and increases the risk of overfitting. It also explains why all models showed unstable behaviour on the minority class.

These results suggest future improvements:

- Applying data augmentation techniques.
- Training for more epochs with early stopping and learning-rate scheduling.
- Class imbalance mitigation strategies
- Use of higher resolution inputs or patch-based methods

9 Code Availability

The source code developed in this project is publicly available and can be accessed at:

https://github.com/pfa13/BreastLesionDetectionMammography

The repository includes data preprocessing scripts, model implementations, training pipelines, and evaluation code, ensuring reproducibility of the experiments.

10 Conclusion

In this work, we evaluated multiple object detection paradigms for breast lesion detection in mammography, including one-stage (YOLO, RetinaNet), two-stage (Faster R-CNN), and transformer-based (RT-DETR) models.

The results showed that all models could detect some masses, but all of them had difficulties with calcifications, which were the most challenging lesions due to their small size and low contrast.

Among the evaluated methods, YOLO achieved the best overall performance and the most balanced results. Faster R-CNN obtained moderate performance and showed potential with more training. RetinaNet was strongly limited by short training time, while RT-DETR detected some masses but struggled with calcifications.

These findings show that both model architecture and training conditions are important in medical imaging tasks.

Future work should focus on larger datasets, longer training, higher-resolution images, and methods specifically designed for small lesion detection.